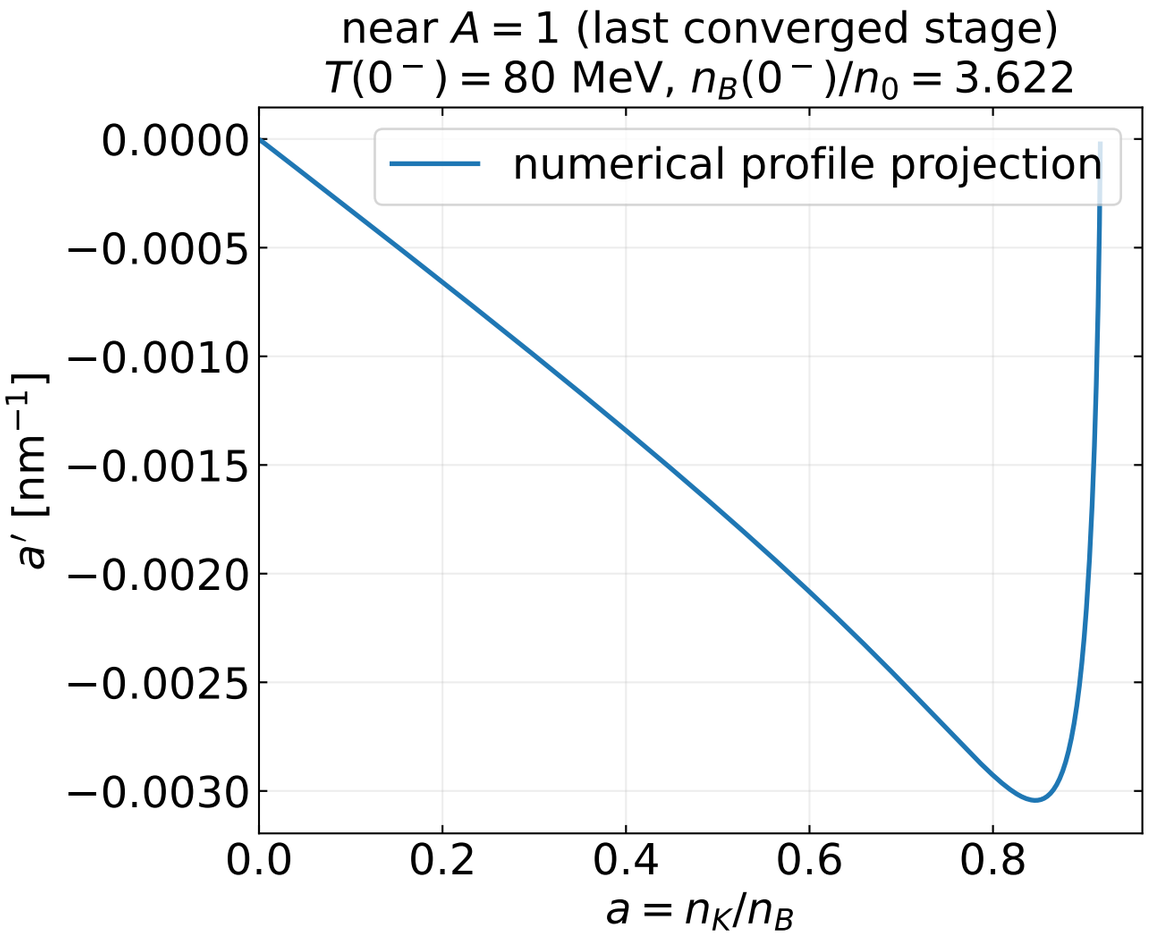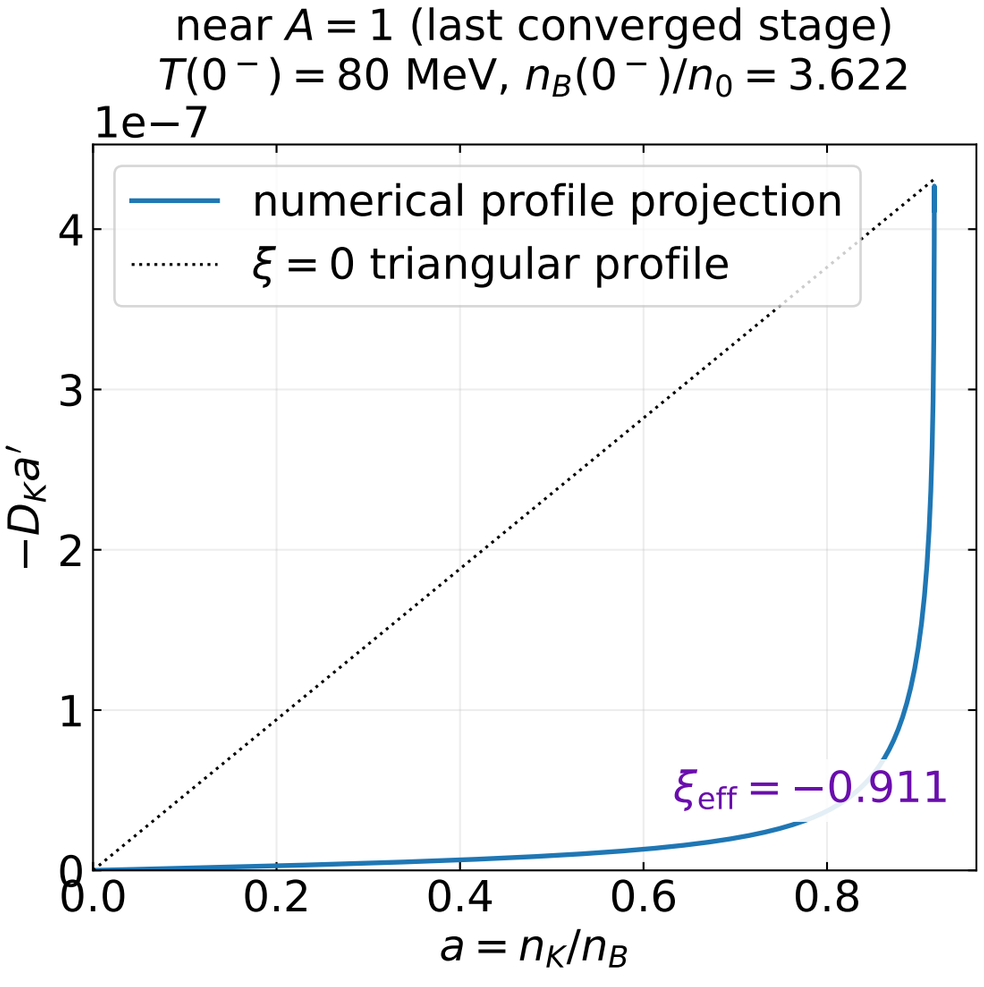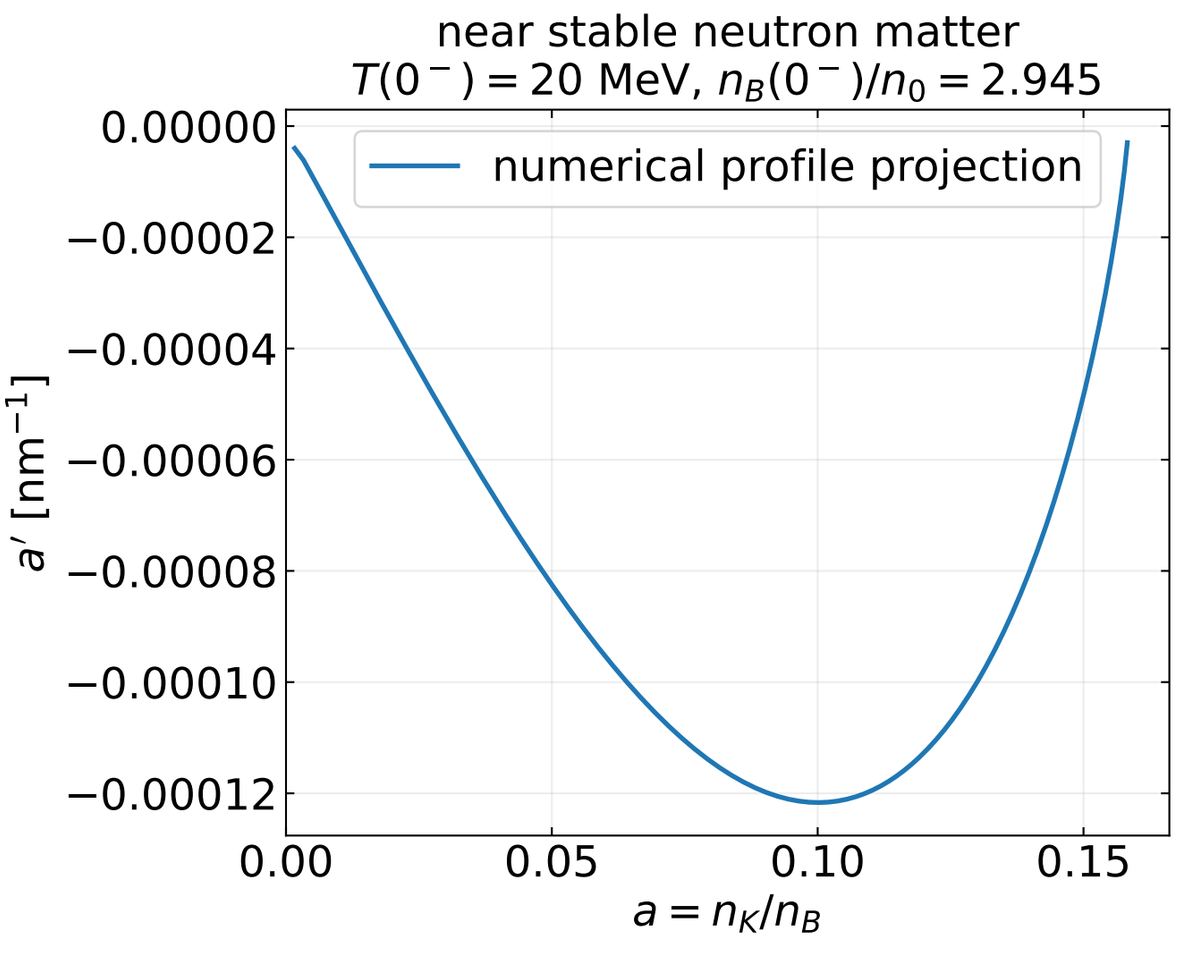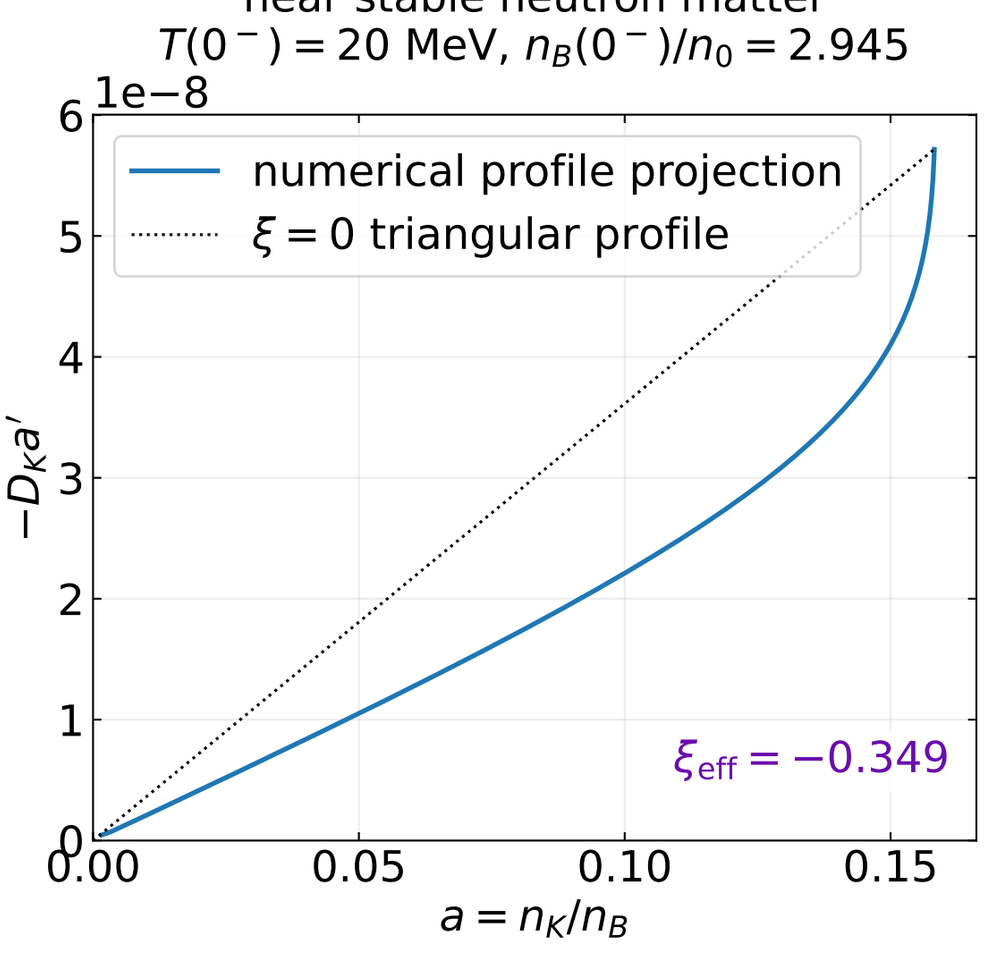Full numerical phase profiles and the $\xi = 0$ raw-flux test